Existence of Every Term of OEIS A396673

Consecutive Odd Squarefree Semiprimes
with Prescribed Prime GCD

Primary author (AI): ChatGPT (OpenAI)

11 November 2026

Abstract

For every odd prime p, there are infinitely many triples of consecutive odd squarefree semiprimes having pairwise greatest common divisor p. In fact, such triples can be chosen consecutive in the complete semiprime sequence, with span bounded by a constant multiple of p. It follows that every term of OEIS A396673 is well-defined. We give the construction explicitly, using the established Maynard–Tao theorem and congruences that exclude unwanted semiprimes. No conjecture about prime pairs or prime tuples is assumed.

Keywords: semiprimes; squarefree integers; prescribed gcd; A396673; Chinese remainder theorem.
OEIS entry: A396673.
MSC 2020: 11N05, 11N36, 11A41.

1 Claim and application

Let $\Omega(n)$ count the prime factors of n with multiplicity. Write

$$\{n \geq 2 : \Omega(n) = 2\} = \{S_1 < S_2 < \cdots\},$$

so prime squares are included. Write $T_1 < T_2 < \cdots$ for the subsequence of odd squarefree semiprimes. Consecutiveness below refers to one of these ordered sequences, not to consecutive integers.

Theorem 1. *There is an absolute constant $H > 0$ such that, for every odd prime p, infinitely many indices k satisfy all of the following:*

1. *S_k, S_{k+1}, S_{k+2} are odd and squarefree;*
2. *$\gcd(S_{k+i}, S_{k+j}) = p$ whenever $0 \leq i < j \leq 2$;*
3. *$S_{k+2} - S_k \leq pH$.*

In particular, these triples are also consecutive members of the sequence (T_j).

Corollary 2 (Well-definedness of A396673)**.** *Let p_n be the n-th prime. For every $n \geq 2$, the set*

$$B_n = \{T_{j+1} : \gcd(T_j, T_{j+1}, T_{j+2}) = p_n\}$$

is infinite. Consequently $A396673(n) = \min B_n$ exists.

Proof. Since p_n is odd for $n \geq 2$, Theorem 1 supplies infinitely many qualifying triples. Pairwise gcd p_n implies common gcd p_n. Distinct triples have distinct middle terms, and every nonempty set of positive integers has a least element. The definition of $\min B_n$ agrees with the OEIS entry [1]. □

2 Established prime-producing input

A finite family of integral linear forms is called *admissible* if, for every prime ℓ, there is a residue class of the variable modulo ℓ on which none of the forms is divisible by ℓ.

Maynard–Tao input. There is an integer K_3 such that every admissible family of $K \geq K_3$ forms

$$An + b_1, \ldots, An + b_K,$$

where A is a positive integer and the b_i are distinct integers, has at least three prime values for infinitely many positive integers n.

This is the common-leading-coefficient case of the Maynard–Tao theorem; see Maynard [2] and the formulation in Banks, Freiberg and Turnage-Butterbaugh [3], Section 1. The threshold K_3 is independent of the fixed coefficients. This established theorem is the analytic input to the proof.

3 Proof of Theorem 1

Fix an odd prime p. Choose a fixed integer $K \geq \max\{3, K_3\}$ and set

$$h_i = (i-1)K! \quad (1 \leq i \leq K), \qquad D = (K-1)K!, \qquad H = D.$$

The shifts $\{h_1, \ldots, h_K\}$ are admissible: modulo a prime $\ell \leq K$ they all vanish, and modulo a prime $\ell > K$ they occupy at most $K < \ell$ residue classes.

Step 1: exclude every noncandidate position. Define the finite set of unwanted offsets

$$\mathcal{T} = \{0, 1, \ldots, pD\} \setminus \{ph_1, \ldots, ph_K\}.$$

For each $t \in \mathcal{T}$, choose a distinct prime $\ell_t > pD + p$. The Chinese remainder theorem gives an integer a satisfying

$$pa + t \equiv 0 \pmod{\ell_t^3} \qquad (t \in \mathcal{T}).$$

These congruences are solvable because p is invertible modulo every ℓ_t^3. Put

$$M = \prod_{t \in \mathcal{T}} \ell_t^3, \qquad u = Mn + a, \qquad L_i(n) = Mn + a + h_i.$$

For every $t \in \mathcal{T}$ and every sufficiently large n,

$$\ell_t^3 \mid pu + t \quad \Longrightarrow \quad \Omega(pu + t) \geq 3.$$

Thus every semiprime in $[pu, p(u + D)]$ must be one of the candidates $pL_i(n)$.

Step 2: verify admissibility after imposing the congruences. If $\ell \mid M$, then $\ell = \ell_t$ for some $t \in \mathcal{T}$, and

$$p(a + h_i) \equiv ph_i - t \pmod{\ell_t}.$$

By construction $ph_i \neq t$ and $0 < |ph_i - t| \leq pD < \ell_t$. Hence no $a + h_i$ is divisible by ℓ_t, so none of the forms $L_i(n)$ vanishes modulo a prime dividing M.

If $\ell \nmid M$, the map $n \mapsto Mn + a$ is a bijection modulo ℓ. The admissibility of the original shifts therefore provides a residue class of n avoiding all zeros of the L_i. This proves admissibility at every prime. The forms have positive leading coefficient M and distinct constant terms, as required.

Step 3: obtain consecutive semiprimes and finish the proof. The Maynard–Tao input now supplies infinitely many n for which at least three of $L_1(n), \ldots, L_K(n)$ are prime. Restrict to sufficiently large such n that every $L_i(n) > p$. List all prime values among these forms as

$$q_1 < q_2 < \cdots < q_r, \qquad 3 \leq r \leq K.$$

For a candidate position, additivity of Ω gives

$$\Omega\big(pL_i(n)\big) = 1 + \Omega\big(L_i(n)\big).$$

Therefore $pL_i(n)$ is a semiprime exactly when $L_i(n)$ is prime. Together with Step 1, this shows that the *only* semiprimes in $[pu, p(u + D)]$ are

$$pq_1 < pq_2 < \cdots < pq_r.$$

They form a consecutive block in the complete semiprime sequence. In particular, its first three terms can be written as

$$S_k = pq_1, \qquad S_{k+1} = pq_2, \qquad S_{k+2} = pq_3.$$

Because $q_i > p \geq 3$, each pq_i is odd and squarefree. The primes q_i are distinct, so

$$\gcd(pq_i, pq_j) = p \gcd(q_i, q_j) = p \qquad (i \neq j).$$

Also

$$S_{k+2} - S_k = p(q_3 - q_1) \leq pD = pH.$$

As the successful arguments n are unbounded, $pq_1 \geq p(Mn + a)$ is unbounded. Hence the construction yields infinitely many distinct triples. Finally, a triple consecutive in the full semiprime sequence and belonging to (T_j) is necessarily consecutive in (T_j). This proves Theorem 1.

4 Scope of the result

The proof establishes existence and infinite recurrence for each prescribed odd prime. It does not locate the first qualifying triple or determine its cofactor gaps. No bound on the exterior semiprime gaps is needed for this application. Although the intermediate construction may contain more than three semiprimes, taking its first three preserves consecutiveness and the gcd property.

The argument is a consequence of established Maynard–Tao and Chinese remainder theorem methods, with the latter adapted to exclude semiprimes by forcing a prime cube divisor. This note does not claim that the well-definedness of A396673 was a previously recognized open problem, or assert independent novelty relative to the literature.

AI authorship and verification disclosure

ChatGPT (OpenAI) is credited as the **primary AI author** of the mathematical drafting and proof presentation. This authorship credit does not imply that the proof or its cited sources have been independently verified. The mathematical claims and references should be checked before publication or reuse.

References

[1] H. Pfoertner, *A396673*, The On-Line Encyclopedia of Integer Sequences, entry created 2 June 2026. https://oeis.org/A396673. Accessed 2 November 2026.

[2] J. Maynard, *Small gaps between primes*, Annals of Mathematics **181** (2015), no. 1, 383–413. doi:10.4007/annals.2015.181.1.7.

[3] W. D. Banks, T. Freiberg and C. L. Turnage-Butterbaugh, *Consecutive primes in tuples*, Acta Arithmetica **167** (2015), no. 3, 261–266. doi:10.4064/aa167-3-4. Also arXiv:1311.7003.